

Community Emergency Response App (CERA)

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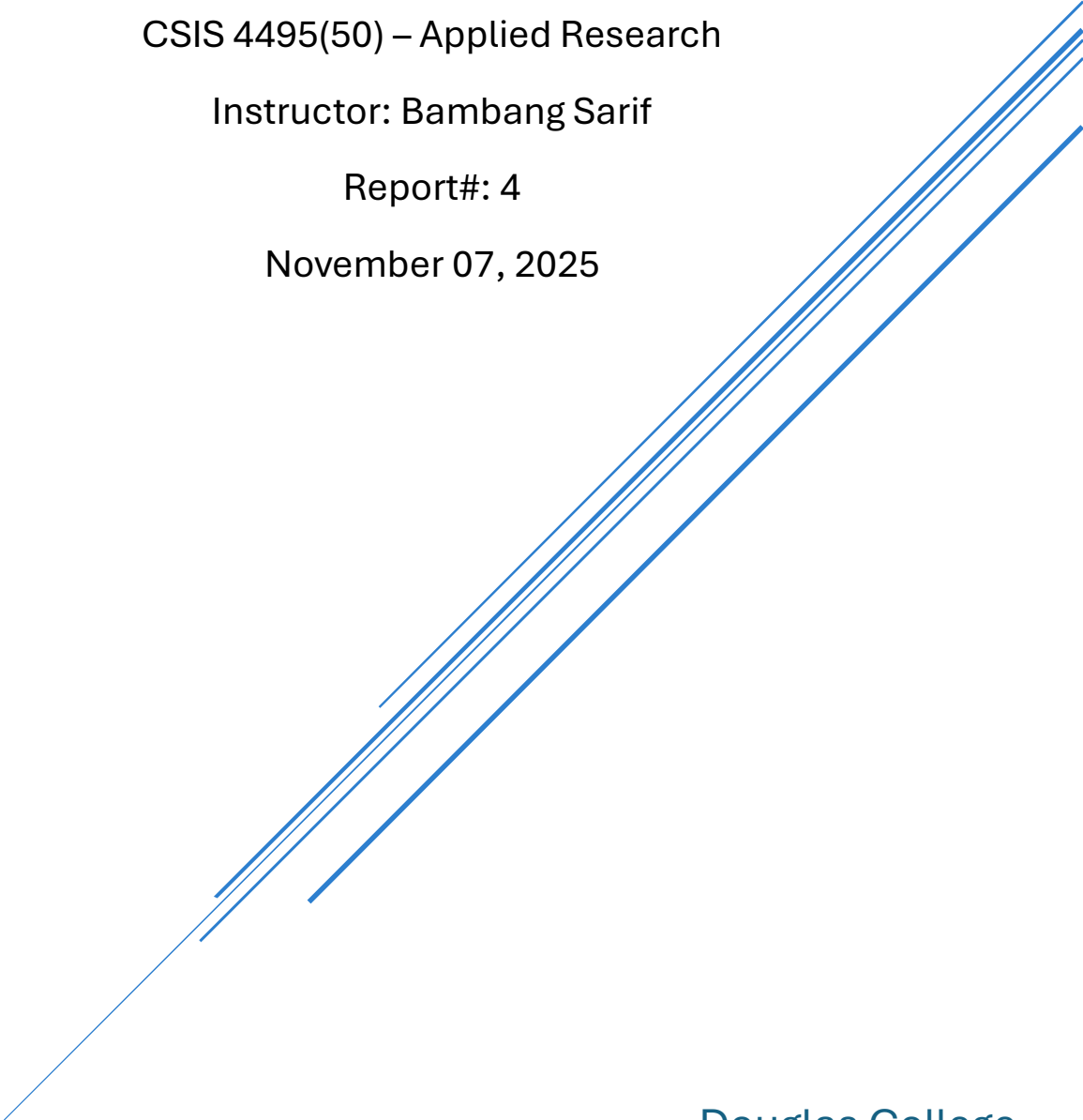
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CSIS 4495(50) – Applied Research

Instructor: Bambang Sarif

Report#: 4

November 07, 2025

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Douglas College
CSIS 4495

Work Logs

Date	Number Of Hours	Description of work done
Oct 25, 2025	3.5	Studied Google Maps API & React Native Maps documentation to understand MapView, Markers, Permissions, Clustering, and Custom Map Styling. Set up initial map test screen.
Oct 26, 2025	2	Integrated live location access using expo-location , requested permissions properly, handled GPS fallback & accuracy levels.
Oct 29, 2025	4.5	Configured MapViewClustering and displayed dynamic incident markers from backend /incidents API. Tested zoom-level-based clustering & UI performance on different devices.
Oct 30, 2025	2	implemented clickable markers → opening incident details modal. Worked on user interaction flow for selecting incident → showing incident metadata (type, severity, reporter).
Oct 31, 2025	4	implemented photo upload during incident report : used ImagePicker → sent multipart/form-data → backend uploads to Cloudinary using Cloudinary SDK and stores returned URL in MongoDB.

Nov 05, 2025	2.5	fixed backend multer directory handling & replaced local storage with Cloudinary-based upload. Removed unused /uploads folder references.
Nov 06, 2025	2	Debugged incident creation failing when image is missing. Added conditional checks and default placeholder images.

Description of Work done

Over the past two weeks, the primary focus was on enhancing the location-based features and supporting media uploads to improve the real-world usability (as told by professor) of the CERA application.

1. Map System & Nearby Incident UI

- Integrated Google Maps (React Native Maps) with clustering for efficient visualization.
- Added user location tracking with permission handling & fallback behavior.
- Implemented dynamic marker rendering, updated automatically based on API results.
- Created incident preview modal triggered by clicking on a marker.
- Added filter controls based on severity levels (Low / Medium / High).

2. Cloudinary Image Upload Integration

- Removed reliance on local filesystem storage.
- Configured backend to use Cloudinary Node SDK for secure image hosting.
- Implemented multipart/form-data upload from Expo image picker.
- Ensured correct DB storage of returned Cloudinary URLs.
- Added error handling for slow network or failed uploads.

3. UI & Navigation Improvements

- Improved theme compatibility for light and dark modes.
- Enhanced user interaction flow for Reporting → Reviewing → Viewing Reports.
- Updated Volunteer map to only show assigned or nearby incidents.

Testing & Debugging

- No tested on physical Android device yet, but tested on iOS emulator to verify location accuracy.
- Validated upload times and image retrieval performance.
- Resolved incorrect marker coordinate scaling and missing reporter field in modal.
- Confirmed **cloud URLs persist correctly** on re-login and across different dashboards.

Repo Check-In

Checked-in folders and file since report 3:

1. Backend:

- /backend/routes/incidents.js- Added Cloudinary upload logic to replace local uploads/ storage.
- /backend/utils/cloudinary.js- Created Cloudinary config helper containing environment-bound credentials, secure URL presets, and reusable upload function
- /backend/models/Incident.js- Updated schema to include photos: [String] with support for multiple uploaded images
- /backend/controllers/incidentController.js- Added modularized controller for image uploads, map-based incident querying, and cloud URL normalization

2. Frontend:

- /app/screens/ReportIncidentScreen.tsx- Integrated Cloudinary upload flow.
- /app/screens/MapScreen.tsx- Implemented MapView + Clustering, added markers for incidents fetched via /incidents API, and modal popup for incident details.
- /app/components/IncidentMarker.js- Extracted marker UI & logic into reusable component for clarity and maintainability.

- /app/components/IncidentDetailModal.js- Created modal UI that displays full incident details (photo, type, severity, reporter, coordinates). Triggered on marker press.

Kanban Board

Epic 5: Maps & Geolocation Integration → Task in review (Map screen integration for the Incident nearby screen with a filter to set the radius to see the incidents near the user)

Epic 5: Maps & Geolocation Integration → Task in review (Cloudinary image cloud storage integration for storage of images posted by the users for incidents)

Images from GitHub Project

Title	Assignee	Status	Priority	Sprint	Estimate	Labels	Linked pull requests
Epic 2: Basic App & Backend Skeleton (Estimate: 17)							
+ Add item							
Epic 3: Registration, Login, Role-Based Access (Estimate: 3)							
12 Frontend Auth Screens #10	Harleen-kaur2002	In Progress	Medium	Sprint 2		enhancement frontend	
13 Backend Auth with JWT #11	Harleen-kaur2002	In Progress	Medium	Sprint 2		backend enhancement	
14 Role-Based Access #12	dappysandhu	Done	Medium	Sprint 2		backend enhancement	
15 Volunteer Approval Flow & Security Check #13		Not Ready	Low	Sprint 2		backend enhancement	
+ Add item							
Epic 4: Incident Reporting & Volunteer Dispatch (Estimate: 12.5)							
16 Incident Reporting & Volunteer Dispatch #35	dappysandhu	Review/Test	High	Sprint 3	8	backend enhancement	#37
17 Not able to fetch nearby incidents after merging to d... #38	dappysandhu	Done	Urgent	Sprint 3	2.5	backend bug	
18 FIX Volunteer Dispatch Screen #42	dappysandhu	Done	Urgent	Sprint 3	2	bug enhancement fix	
+ Add item							
Epic 5: Maps & Geolocation Integration (Estimate: 18)							
19 Implement 'Nearby Incidents Map + Improved Incid... #45	dappysandhu	Review/Test	High	Sprint 4	12	backend enhancement	
20 Integrate Cloudinary Image Upload for Incident Repo... #47	dappysandhu	In Progress	High	Sprint 4	6	backend enhancement	
+ Add item							

Figure 1: Kanban Board showing tasks in Backlog, In Progress, and Done.

Community Emergency Response App

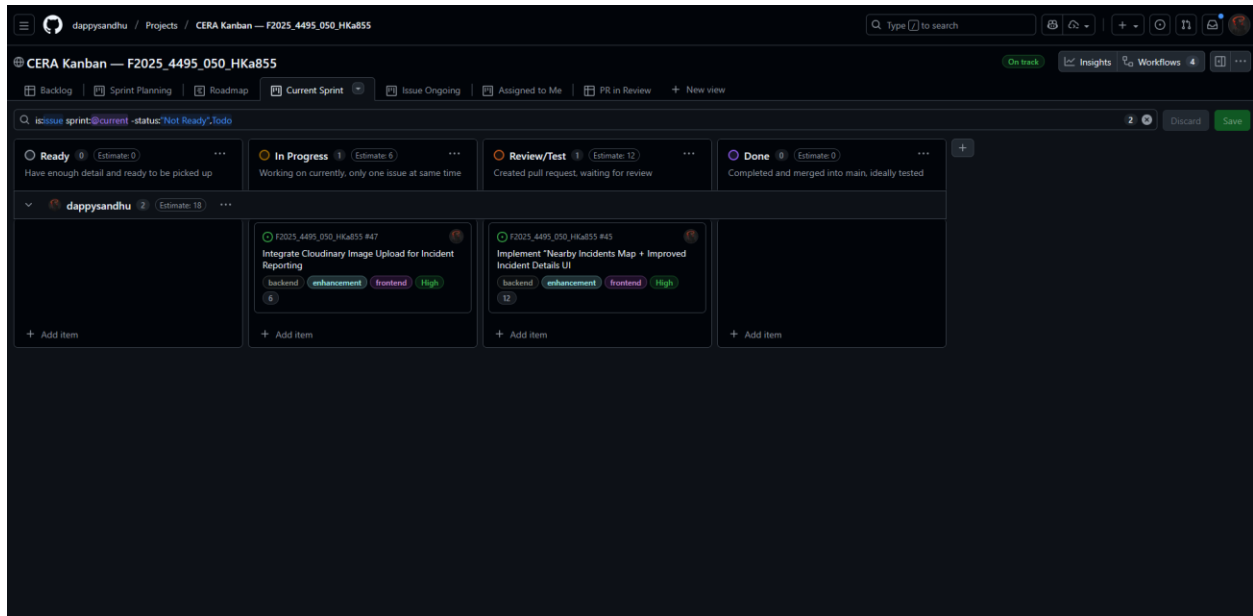


Figure 2: Current Ongoing Sprint.

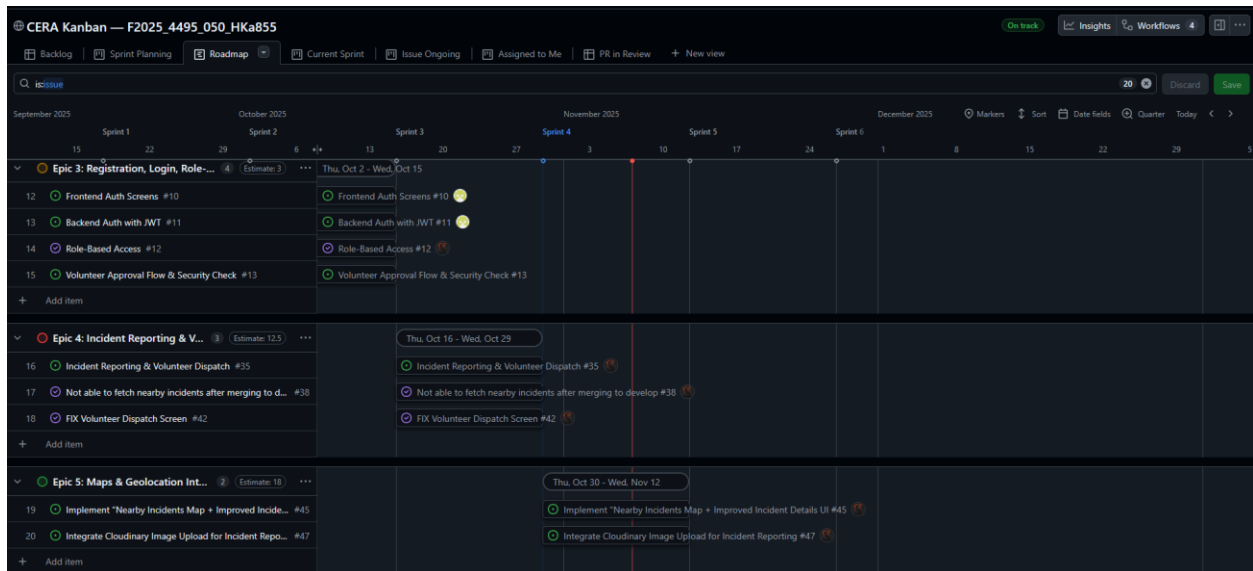


Figure 3: Project Roadmap.

Screens Implemented according to above tasks:

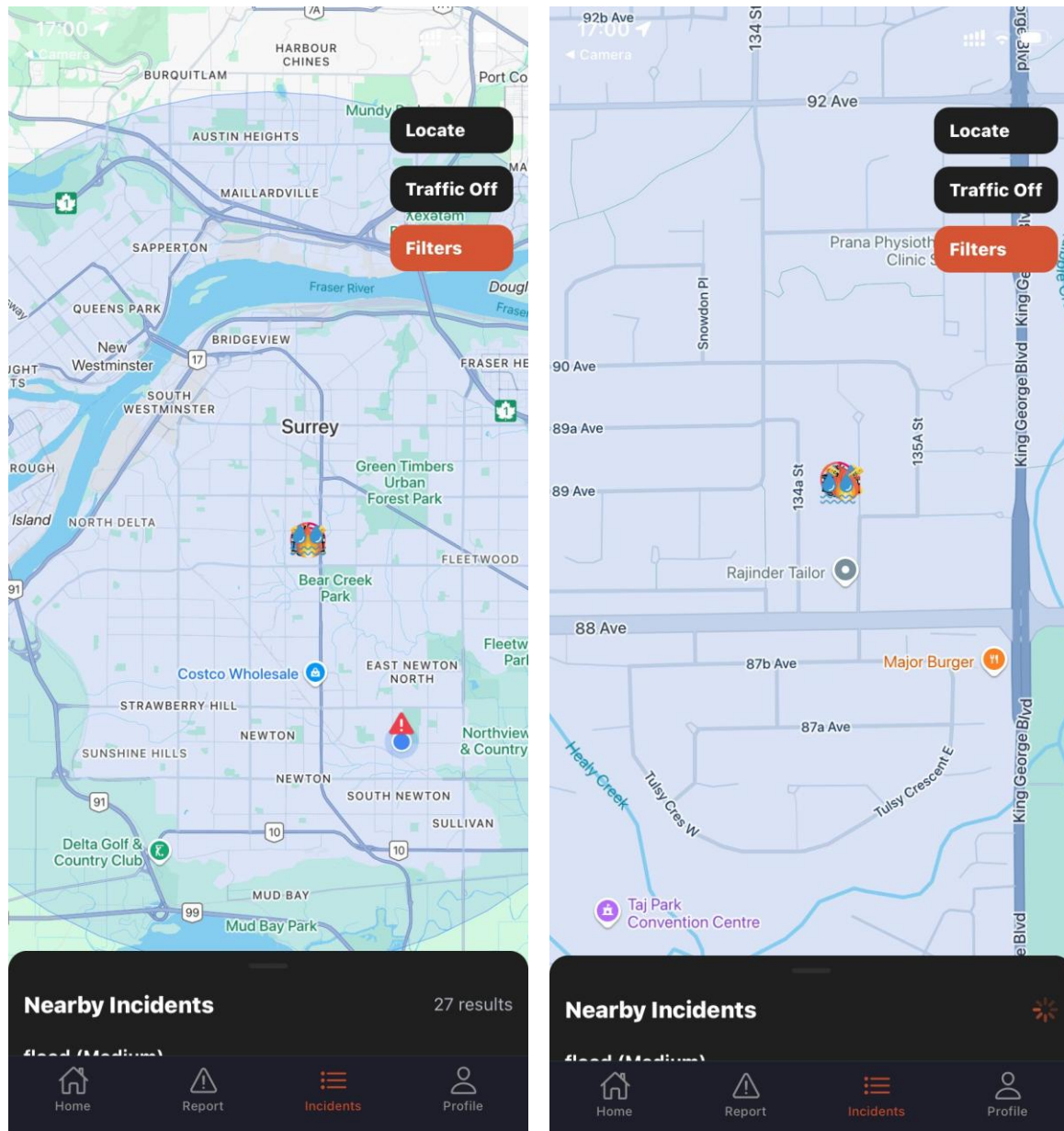


Figure 4. Map with all incidents

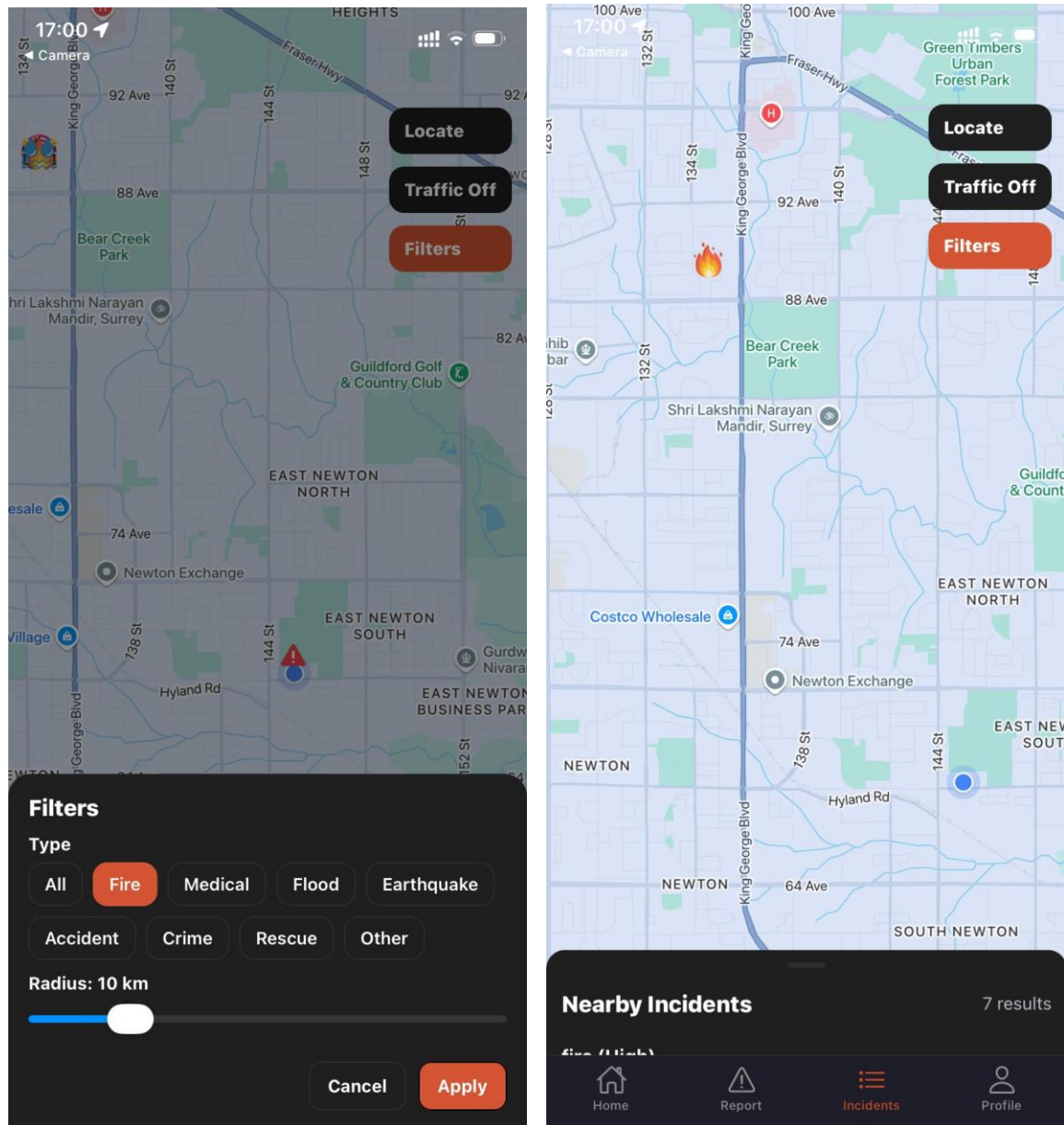


Figure 5: Incidents nearby with Fire filter

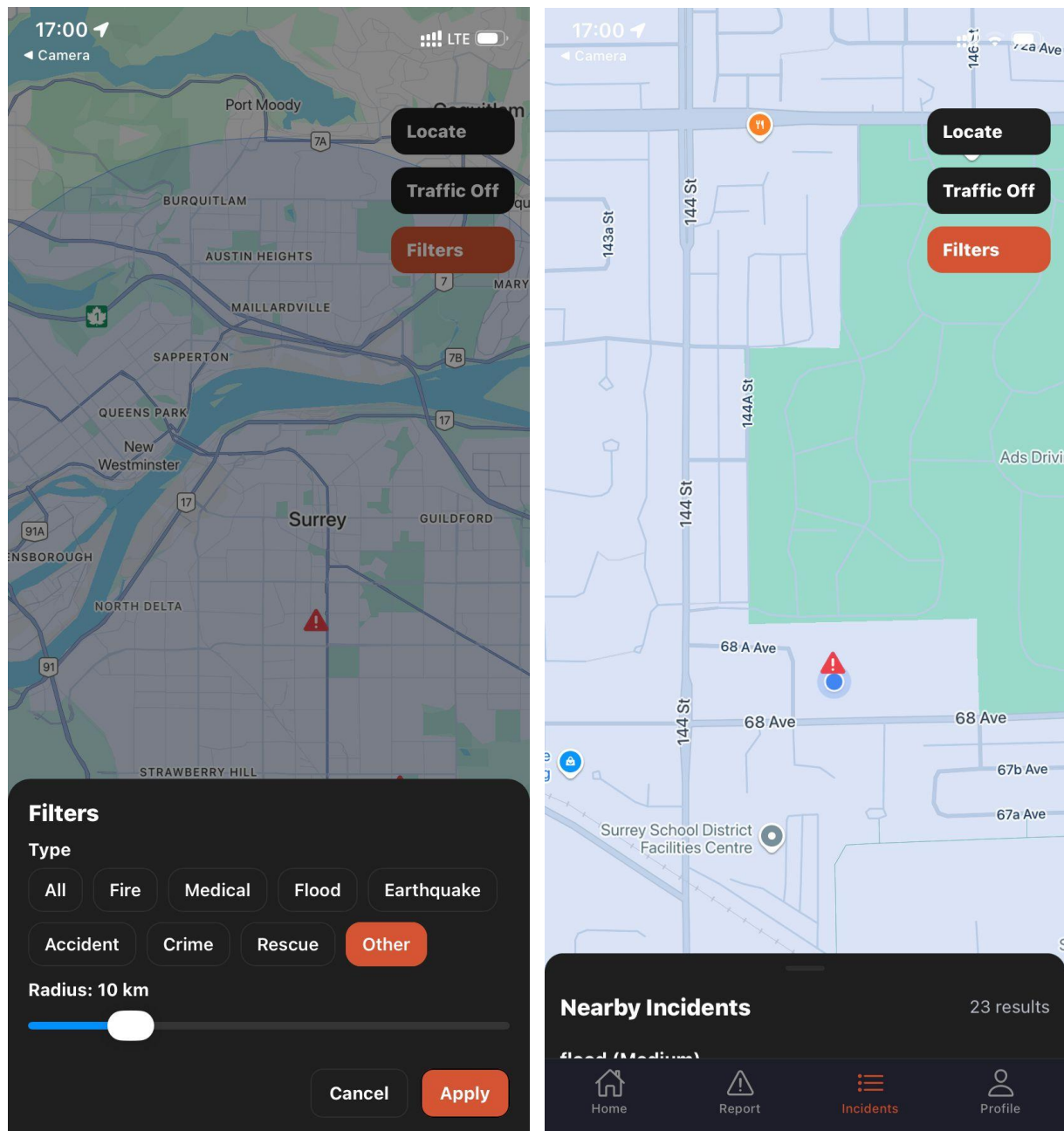


Figure 6: Incidents nearby with “other” filter

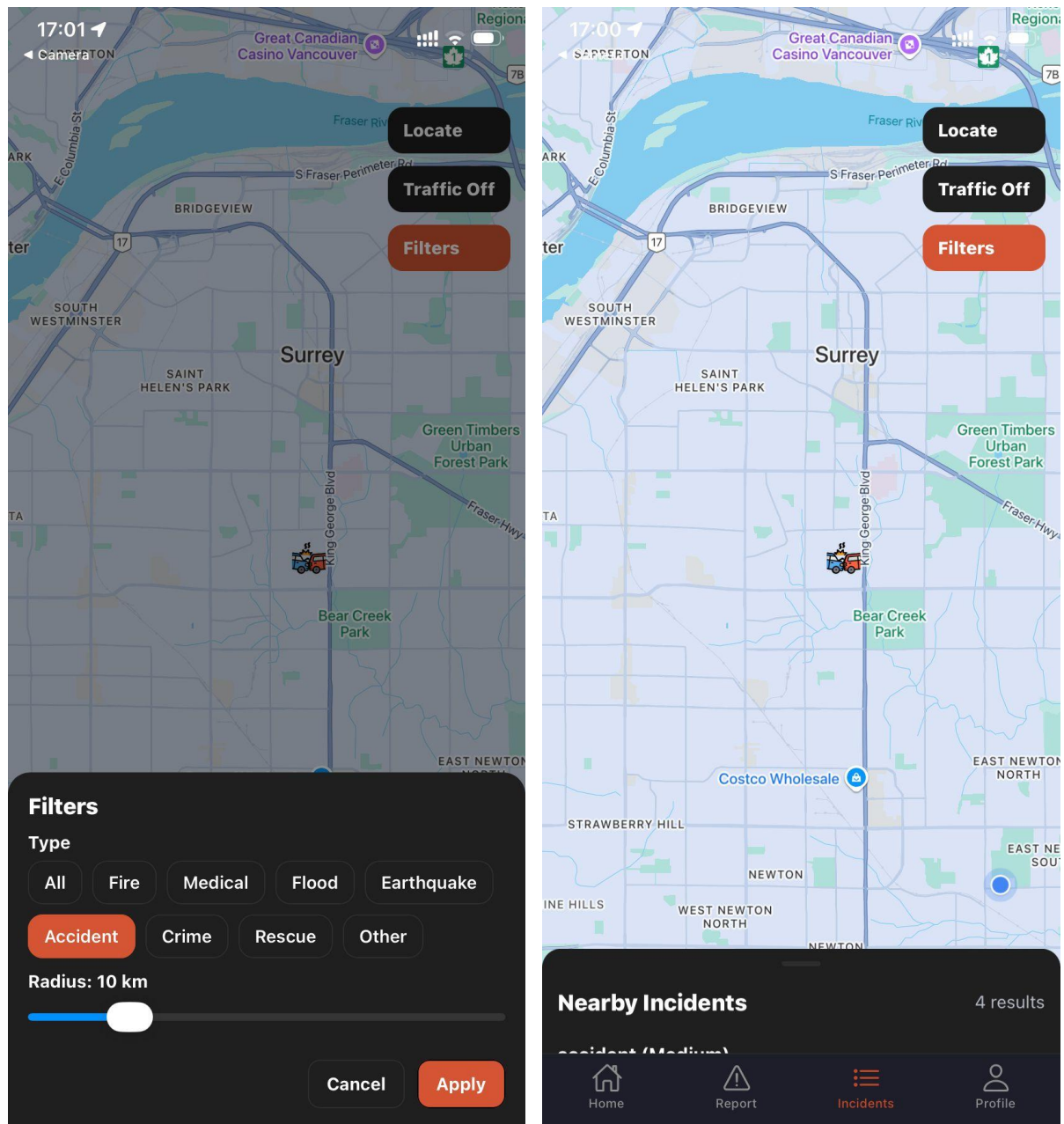


Figure 7: Incidents nearby with “accidents” filter